Bayesian Analysis of Repeated-Measures Designs

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1. Data description

The within-subject data (repeated measures) are from a survey study examining the effect of intermittent fasting (IF) on the quality of life (QoL) scores of 15 participants, each of whom once answered the same 30-question questionnaire before, during, and after the intervention. The data set (long format) consists of seven columns:

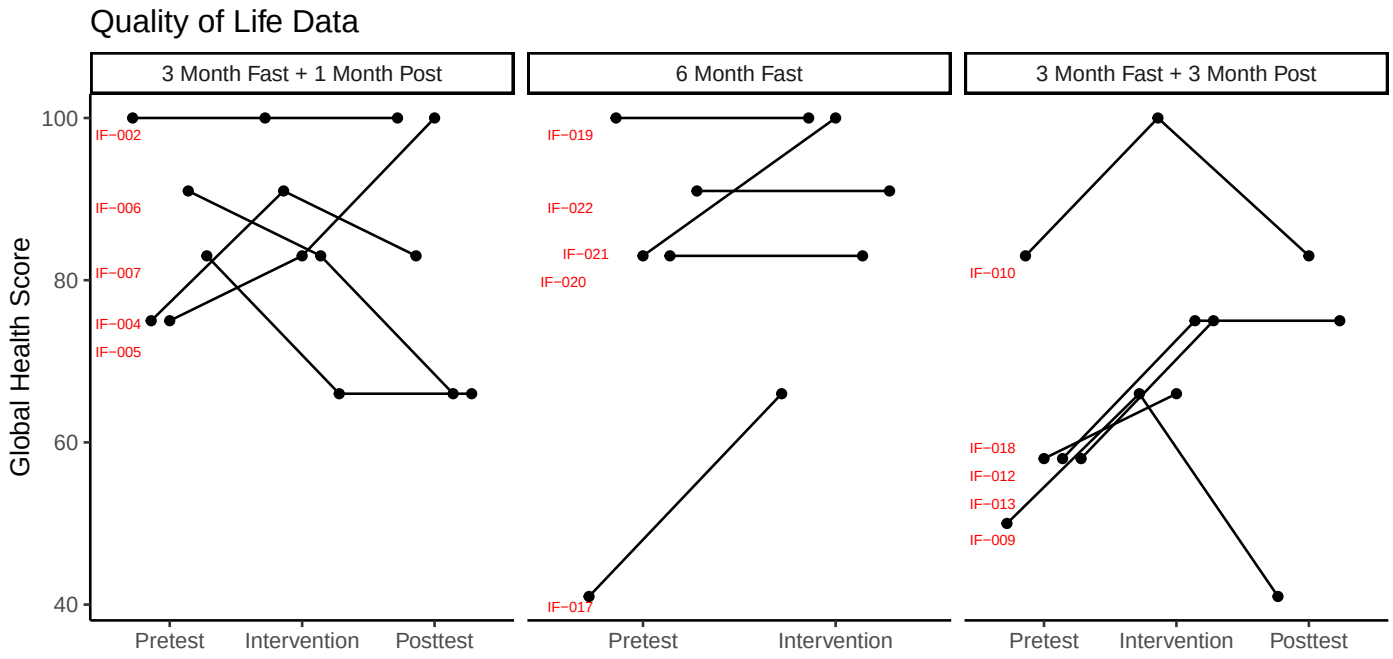
- **ID** - [factor] participant identifier; 15 levels;
- **Phase** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest;
- **Fast** - [numeric] fasting group identifier;

  0 as 3 months fasting and 1 month follow-up,
  1 as 6 months fasting,
  and 2 as 3 months fasting and 3 months follow-up;

- **Total** - [numeric] total QoL score;
- **Global** - [numeric] global health score;
- **Physical** - [numeric] physical function score;
- **Fatigue** - [numeric] fatigue score.

```
invisible(lapply(c("ggplot2", "ggrepel", "BayesFactor"),
                 library, character.only=T)) #load packages

df <- read.csv("~/Downloads/QoL_v1.1.csv", stringsAsFactors=T)
df$Phase <- factor(df$Phase, labels=c("Pretest", "Intervention", "Posttest"))
df$Fast <- factor(df$Fast, labels=c("3 Month Fast + 1 Month Post",
                                    "6 Month Fast",
                                    "3 Month Fast + 3 Month Post"))
df <- na.omit(df)
ggplot(df, aes(x=Phase, y=Global, group=ID))+
  geom_point(position=position_dodge(.7))+
  geom_line(position=position_dodge(.7))+ #jitter to avoid overplotting
  geom_text_repel(data=subset(df,Phase=="Pretest"), aes(label=ID),
                  vjust=2, hjust=2, segment.color=NA, size=2, col="red")+
  xlab("")+ylab("Global Health Score")+
  ggtitle("Quality of Life Data")+
  facet_wrap(~Fast, scales="free_x")+
  theme_classic()
```



2. Frequentist: Repeated-measures analysis of variance

A one-way repeated-measures analysis of variance (RM-ANOVA) is a statistical procedure used to determine any significant differences among multiple sets of data measurements (i.e., levels of a single factor), where all measurements are taken on the same set of subjects. Unlike the standard one-way ANOVA, RM-ANOVA can account for the dependence between measurements by further partitioning the within-groups variability into the subject variability `Error: ID` and error variability `Error: ID:Phase`.

Traditional RM-ANOVA requires a **complete case**, which needs to be measured for all conditions in each subject. However, IF-012 and 018 contain missing values for the posttest.

Scenario 1: exclude all the subjects who missed the posttest, run omnibus RM-ANOVA for the remaining eight subjects with three conditions, check for the sphericity/circularity assumption, and may conduct the post-hoc analysis.

Scenario 2: run RM-ANOVA for all 15 subjects with two conditions (pretest and intervention).

Scenario 1

```
colnames(df)[which(names(df)=="Global")] <- "Response" #set the response variable
dat <- droplevels(subset(df, !(ID %in% paste0("IF-0", c(12,17,18,19,20,21,22)))))
# summary(aov(Response ~ Phase + Error(ID/Phase), data=dat)) #RM-ANOVA
fit <- nlme::lme(fixed=Response ~ Phase,
                 random=~1 | ID, #random intercept for each subject
                 data=dat, method="ML")
summary(fit)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   200.9194 206.8097 -95.45971
```

```
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept) Residual
## StdDev:    12.31319 9.599533
##
## Fixed effects:  Response ~ Phase
##                     Value Std.Error DF   t-value p-value
## (Intercept)       76.875  5.901158 14 13.027105  0.0000
## PhaseIntervention  6.125  5.131166 14  1.193686  0.2524
## PhasePosttest     -0.125  5.131166 14 -0.024361  0.9809
##  Correlation:
##                   (Intr) PhsInt
## PhaseIntervention -0.435
## PhasePosttest     -0.435  0.500
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -1.42504332 -0.56877444 -0.07821163  0.58233387  1.80480997
##
## Number of Observations: 24
## Number of Groups: 8
```

```
nlme::intervals(fit) #95% confidence intervals for the fixed effects
```

```
## Approximate 95% confidence intervals
##
##  Fixed effects:
##                         lower   est.    upper
## (Intercept)         65.035718 76.875 88.71428
## PhaseIntervention   -4.169476  6.125 16.41948
## PhasePosttest      -10.419476 -0.125 10.16948
##
##  Random Effects:
##   Level: ID
##                    lower     est.    upper
## sd((Intercept)) 6.80218 12.31319 22.28912
##
##  Within-group standard error:
##     lower      est.     upper
##  6.788561  9.599533 13.574459
```

TL;DR, the below Mauchly's sphericity test passed, and RM-ANOVA may be valid to run.

► The p-value of .403 for the conditions is not statistically significant at an alpha level of .05 ($F(2, 14) = .97$). Hence, fail to reject the null hypothesis that the mean responses over three phases are the same. According to the analysis of eight subjects, the mean global health scores do not change before, during, or after the intervention.

```
ez::ezANOVA(data=dat, wid=ID, within=.(Phase), dv=Response, type=3) #Mauchly's sphericity test
```

```
## $ANOVA
##   Effect DFn DFd         F         p p<.05        ges
## 2  Phase   2  14 0.9697055 0.4032648       0.03373454
##
## $`Mauchly's Test for Sphericity`
##   Effect         W         p p<.05
## 2  Phase 0.8855019 0.6943342
##
## $`Sphericity Corrections`
##   Effect       GGe     p[GG] p[GG]<.05      HFe     p[HF] p[HF]<.05
## 2  Phase 0.8972649 0.3967539           1.186851 0.4032648
```

Scenario 2

```
dat2 <- droplevels(subset(df, Phase!="Posttest"))
# summary(aov(Response ~ Phase + Error(ID/Phase), data=dat2)) #RM-ANOVA, same as below
fit2 <- nlme::lme(fixed=Response ~ Phase,
                  random=~1 | ID, #random intercept for each subject
                  data=dat2, method="ML")
summary(fit2)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat2
##        AIC      BIC   logLik
##   245.7699 251.3747 -118.885
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept) Residual
## StdDev:    13.36165 7.909765
##
## Fixed effects:  Response ~ Phase
##                       Value Std.Error DF   t-value p-value
## (Intercept)       75.26667  4.149853 14 18.137187  0.0000
## PhaseIntervention  7.73333  2.989610 14  2.586736  0.0215
##  Correlation:
##                   (Intr)
## PhaseIntervention -0.36
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -1.65080347 -0.40014013 -0.05744753  0.60509169  1.47613322
##
## Number of Observations: 30
## Number of Groups: 15
```

```
nlme::intervals(fit2) #95% confidence intervals for the fixed effects
```

```
## Approximate 95% confidence intervals
##
##  Fixed effects:
```

```
##                       lower      est.    upper
## (Intercept)       66.66792 75.266667 83.86542
## PhaseIntervention  1.53868  7.733333 13.92799
##
##  Random Effects:
##   Level: ID
##                     lower     est.    upper
## sd((Intercept)) 8.733821 13.36165 20.44164
##
##  Within-group standard error:
##     lower      est.     upper
##  5.530399  7.909765 11.312815
```

RM-ANOVA result should be consistent with the linear mixed-effects regression `lme`.

► The p-value of .0215 for the conditions is statistically significant at an alpha level of .05 ($F(1, 14) = 6.691 = 2.586736^2 = t^2_{14}$). Hence, reject the null hypothesis. The mean global health scores increase after receiving IF treatment, according to the analysis of all 15 subjects.

3. Bayesian: Using the "*BayesFactor*" package (v 0.9.12-4.6)

The Bayes factor (BF) quantitatively measures the evidence for one statistical model versus another. In the case of the one-way linear mixed-effects model, we compare the full model (which assumes an effect) to the null model (which assumes no such effect). The Bayes factor value of about 3 is moderate evidence for an effect.

```r
#Scenario 1
set.seed(277); anovaBF(Response ~ Phase + ID, whichRandom="ID", data=dat, progress=F)
```

```
## Bayes factor analysis
## --------------
## [1] Phase + ID : 0.4237035 ±0.47%
##
## Against denominator:
##   Response ~ ID
## ---
## Bayes factor type: BFlinearModel, JZS
```

```r
#Scenario 2
set.seed(277); anovaBF(Response ~ Phase + ID, whichRandom="ID", data=dat2, progress=F)
```

```
## Bayes factor analysis
## --------------
## [1] Phase + ID : 2.654517 ±0.72%
##
## Against denominator:
##   Response ~ ID
## ---
## Bayes factor type: BFlinearModel, JZS
```

► The data from eight subjects are $1/0.4237035 \approx 2.360141$ times more likely under the null model than they are under the full model, indicating anecdotal evidence of absence.

► The data from all 15 subjects are 2.654517 times more likely under the full model than they are under the null model, indicating anecdotal evidence for an effect.

4. Bayesian: Within-subject credible interval

Wei, Nathoo, and Masson (2023) proposed Bayesian credible intervals (CrI) for within-subject designs. The motivation is to remove the irrelevant between-subjects variability and reveal the actual pattern of the population means in within-subject designs.

```
install.packages("rmBayes", type="binary") #need R version 4.0.1 or later
```

```
library(rmBayes) #v 0.1.15; rstan v 2.32.5; macOS 13.6.3 M2
rmHDI(dat, whichSubject="ID", whichLevel="Phase", seed=277)
```

```
## $HDI
##                 lower    upper
## Intervention 74.34125 89.68174
## Posttest     69.39069 84.73118
## Pretest      69.45653 84.79702
##
## $`posterior means`
## Intervention     Posttest      Pretest
##     82.01150     77.06093     77.12677
##
## $width
## [1] 7.670245
```

```
rmHDI(dat2, whichSubject="ID", whichLevel="Phase", seed=277)
```

```
## $HDI
##                 lower    upper
## Intervention 78.03401 87.36341
## Pretest      70.93863 80.26803
##
## $`posterior means`
## Intervention      Pretest
##     82.69871     75.60333
##
## $width
## [1] 4.664703
```

5. Summary

Change `colnames(df)[which(names(df)=="NAME")] <- "Response"` to set a new response variable and rerun the analyses.

Global	Scenario 1		Scenario 2	
RM-ANOVA	$p = .40$, $F(2, 14) = .97$	Not significant.	$p = .022$, $F(1, 14) = 6.69$	Significant at $\alpha = .05$.
BF_{10}	0.42	Anecdotal evidence of absence.	2.65	Anecdotal evidence for an effect.
95% CrI	Pretest:[69.46, 84.80] Intervention: [74.34, 89.68] Posttest: [69.39, 84.73]	32% separation, given $n = 8$.	Pretest: [70.94, 80.27] Intervention: [78.03, 87.36]	76% separation, given $n = 15$.

Total	Scenario 1		Scenario 2	
RM-ANOVA	$p = .18$, $F(2, 14) = 1.93$	Not significant.	$p = .0098$, $F(1, 14) = 8.94$	Significant at $\alpha = .01$.
BF_{10}	0.69	Anecdotal evidence of absence.	4.50	Moderate evidence for an effect.
95% CrI	Pretest:[51.70, 55.57] Intervention: [49.88, 53.75] Posttest: [51.49, 55.37]	47% separation, given $n = 8$.	Pretest: [49.35, 51.46] Intervention: [47.50, 49.61]	87% separation, given $n = 15$.

Physical	Scenario 1		Scenario 2	
RM-ANOVA	$p = .11$, $F(2, 14) = 2.63$	Not significant.	$p = .012$, $F(1, 14) = 8.36$	Significant at $\alpha = .05$.
BF_{10}	0.98	Inconclusive.	4.19	Moderate evidence for an effect.
95% CrI	Pretest:[78.12, 89.57] Intervention: [85.08, 96.54] Posttest: [80.99, 92.45]	61% separation, given $n = 8$.	Pretest: [84.41, 92.28] Intervention: [91.17, 99.04]	86% separation, given $n = 15$.

Fatigue	Scenario 1		Scenario 2	
RM-ANOVA	$p = .38$, $F(2, 14) = 1.02$	Not significant.	$p = .11$, $F(1, 14) = 2.98$	Not significant.
BF_{10}	0.43	Anecdotal evidence of absence.	0.92	Inconclusive.
95% CrI	Pretest:[24.87, 47.96] Intervention: [16.09, 39.19] Posttest: [20.46, 43.56]	38% separation, given $n = 8$.	Pretest: [21.66, 34.93] Intervention: [15.05, 28.33]	50% separation, given $n = 15$.

References

Wei, Z., Nathoo, F. S., & Masson, M. E. J. (2023). Investigating the relationship between the Bayes factor and the separation of credible intervals. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-023-02295-1

Wei, Z., Nathoo, F. S., & Masson, M. E. J. (2022). *rmBayes: Performing Bayesian inference for repeated-measures designs*. R package version 0.1.15. https://cran.r-project.org/package=rmBayes